

# PAPER\_360 — J1610+1811 High-z Quasar Jet at z=6.5: Relativistic Lorentz Factor k\_rel Coupling

**Whitepaper Series:** Star-Magic UQFF Phase 2

**Session:** 97

**Source:** gok\_share\_31b5c807a4.txt (Supplemental Gap Analysis Block)

**Classification:** FIRST UQFF high-z quasar jet (z=6.5) with Lorentz factor  $k_{\text{rel}} = \Gamma^2$  relativistic coupling

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## Abstract

J1610+1811 is a blazer-class quasar at  $z = 6.5$  (lookback time  $\sim 12.9$  Gyr) with a relativistic jet Lorentz factor  $\Gamma \approx 4.5$ . UQFF introduces a relativistic coupling constant  $k_{\text{rel}} = \Gamma^2 = k_{\text{rel},0} \times 20.25$  (Lorentz factor squared) to scale the vacuum buoyancy force in the jet frame. The Friedmann Hubble parameter  $H(z = 6.5)$  is computed from  $H(z) = H_0 \sqrt{0.3(1+z)^3 + 0.7}$ , and  $F_{U\_Bi\_i} \approx -8.32 \times 10^{217}$  N is evaluated in the observer frame.

## 2. Core Physics

### 2.1 Relativistic UQFF Coupling

$$k_{\text{rel}} = \Gamma^2 = k_{\text{rel},0} \times \Gamma^2 = k_{\text{rel},0} \times (4.5)^2 = 20.25 \cdot k_{\text{rel},0}$$

The Lorentz factor squared enhancement represents the relativistic Doppler amplification of the vacuum buoyancy force in the jet frame.

### 2.2 Friedmann H(z) at z = 6.5

$$H(z) = H_0 \sqrt{0.3(1+z)^3 + 0.7}$$

For  $z = 6.5$ :

$$(1 + 6.5)^3 = 7.5^3 = 421.875$$

$$H(6.5) = H_0 \sqrt{0.3 \times 421.875 + 0.7} = H_0 \sqrt{126.56 + 0.7} = H_0 \sqrt{127.26}$$

$$H(6.5) \approx 11.3 H_0 = 11.3 \times 67.4 \text{ km/s/Mpc} \approx 761 \text{ km/s/Mpc}$$

### 2.3 Relativistic F\_U\_Bi\_i

$$F_{U\_Bi\_i}^{\text{relativistic}} = F_{U\_Bi\_i}^{\text{standard}} \cdot k_{\text{rel}} = -8.32 \times 10^{217} \times 20.25 \approx -1.69 \times 10^{219} \text{ N}$$

(in the jet co-moving frame; observer-frame is un-boosted)

### 2.4 Lookback Time at z = 6.5

$$t_{\text{lookback}} \approx 12.9 \text{ Gyr}$$

The Universe was only  $\sim 820$  Myr old when this jet was active — the UQFF vacuum energy density was  $\rho_{\text{vac}}(z=6.5) = \rho_{\text{vac},0} \cdot (1+z)^\alpha$ , higher than today by the UQFF expansion index  $\alpha$ .

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### 3. Key Values

| Quantity                          | Formula                 | Value                             |
|-----------------------------------|-------------------------|-----------------------------------|
| $z$                               | Photometric/spectro     | 6.5                               |
| $\Gamma$                          | VLBI jet model          | 4.5                               |
| $k_{\text{rel}}$                  | $\Gamma^2$              | 20.25                             |
| $H(z=6.5)$                        | Friedmann               | $\sim 761 \text{ km/s/Mpc}$       |
| $F_{\text{U\_Bi\_i}}$ (obs frame) | Standard                | $-8.32 \times 10^{217} \text{ N}$ |
| $F_{\text{U\_Bi\_i}}$ (jet frame) | $\times k_{\text{rel}}$ | $-1.69 \times 10^{219} \text{ N}$ |

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### 4. Physical Significance

J1610+1811 at  $z = 6.5$  presents UQFF at the earliest cosmic epoch in the dataset. The  $k_{\text{rel}} = \Gamma^2$  coupling is the first relativistic enhancement factor in UQFF AGN physics — it predicts that high- $\Gamma$  relativistic jets experience systematically larger UQFF vacuum buoyancy forces in their rest frame. This has cosmological implications: early universe ( $z > 5$ ) AGN jets would have experienced larger vacuum buoyancy during the epoch of reionization, potentially accelerating the growth of early massive black holes — addressing the “first quasar” problem.

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### 5. Deduplication Note

- **vs. PAPER\_346-350 (low- $z$  AGN):** All earlier AGN papers in this series have  $z < 1.5$ ; J1610 at  $z = 6.5$  is  $6.5\times$  higher redshift.
  - **vs. PAPER\_360 vs.  $k_{\text{rel}}$ :** No earlier UQFF paper includes the Lorentz factor  $\Gamma^2$  relativistic boost.
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### 6. Classification

**Physics Territory:** FIRST UQFF high- $z$  quasar jet ( $z=6.5$ ) with  $\Gamma^2$  relativistic coupling and Friedmann  $H(z)$

**Scale:** Cosmological ( $z = 6.5$ , lookback  $\sim 12.9$  Gyr)

**CP Implementation:** J1610HighZQuasarJetFUBiCalculator (CondensedPhysics4.py, Session 97)